

CHE 372 Chemical Reactor Analysis & Design

Spring 2026

Course description: This course covers the fundamentals of analysis and design of chemical reactors. The ideas and methods covered in this course have wide application in petrochemical and pharmaceutical processes. Recently, these methods have wide application in modeling biological activities that occur inside the cell. Further this course combines the ideas learned in thermodynamic, transport phenomena and mass & energy balance courses.

Prerequisites: ChE 322 (Thermodynamics), ChE 348 and ChE 354.

Instructor: Thomas Edison,

Office: CPE 4.458, Office Phone: 471-4438

Office Hours: Tu, Th 11:00 - 2:00

Teaching Assistant:

Office Hours:

Class Meeting: Tu,Th 9:30 - 11:00 PM (CPE 2.214)

Recitation: W 12:00 – 1:00 PM (CPE 2.214)

Optional Text:

J.B. Rawlings and J.G. Ekerdt, *Chemical Reactor Analysis and Design Fundamentals*, Second edition, Nob Hill Publishing, 2012.

H. Fogler, *Elements of Chemical Reaction Engineering*, sixth edition, Pearson, 2020.

Final Grade:

Pre-Lecture problems	5 %
In-class Problems	5 %
Homework	5 %
Quizzes	15 %
Project (ASPEN & MATLAB)	10 %
Three Midterm Exams	60 %
Final Exam	(Makeup exam-1 and exam-2)

Final letter grade will be given based on Class average. The class average for this course is B-. There are two requirements for a passing grade C-: You should have an exam grade of 60 points and have a course average that will be determined from your class average and standard deviation.

Homework:

Homework will be assigned once a week and will be due next week. Only selected problems will be graded. Encouraged to use MATLAB.

Pre-Lecture Problems:

Before each lecture students are encouraged to complete a short problem. Answers to pre-class problems has to be hand written (if you use your IPAD you can print) will be collected at the beginning of each lecture.

Lectures and in-class problems: During lecture I will go over the concepts and then I will go over some example problems. Students are encouraged to complete these in class problems before the end of each week and upload it to canvas.

Exams:

There will be three exams during the semester. Exams are closed notes and equation sheet will be provided. There is no cumulative final exam for this course. During final exams week there will be make-up exams 1 and 2.

Exam – I Feb 12th (Evening 7 to 10 pm CPE 2.210, 2.220, 2.216)

Exam – II Mar 12th (Evening 7 to 10 pm CPE 2.210, 2.220, 2.216)

Exam – III Apr 24th (Evening 6 to 9 pm CPE 2.210, 2.220, 2.216)

(You have one week after the exam is returned to you to request for re-grading. Re-grading requests should be submitted in writing.)

Make-up Exams: If you have less than 60% average in the three midterm exams, you have a final chance to pass the course by taking make-up exams. There make-up exams are two hours long. It will be given on April 29th from 2 to 4 pm and May 3rd from 2 to 4 pm. In order to take the makeup exams, you should have at least 40% in the three exams. This is to discourage students from skipping midterm exams.

Cell phones and smart watches during the exams: At the beginning of exam, you have to switch of the cell phone and place it in your back pack. You may not wear smart watches during the exam.

Bathroom breaks during the last one-hour exam is not encouraged. Also, one student at a time is allowed to go to the bathroom closes to the exam room.

The University of Texas at Austin provides upon request appropriate academic accommodations for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-6441 TTY.

Absence: No excuse for missed tests, exams, or quizzes will be accepted other than written certified medical excuses. If you miss an exam due to health or family reason, you have to take the final exam. Final exam will also have problems from missed exam.

Academic Integrity:

The University has a strict policy on academic integrity. Any form of plagiarism or academic dishonesty will NOT be tolerated in ChE 372. If you have any questions, please consult the websites below:

http://deanofstudents.utexas.edu/sjs/acint_student.php

http://www.utexas.edu/courses/clubmed/plgrism_cit.html

In particular, students are obliged to follow the Student Honor Code that states “As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity.”

Failure to comply with the University’s policy will result in a “zero” in the assignment and an “F” for CHE 372.

Note: During the lecture period you may sit where you feel comfortable. If any groups of students constantly talk and disturb others during lectures, then the instructor has the right to assign seating arrangement for the benefit of the class.

Lecture Outline: Following topics will be covered during the lectures

1. Introduction and review of stoichiometry of reactions. Concept of extent of reactions, stoichiometric matrix, linear independence of chemical reactions (Chapter 2)
2. Review of thermodynamics. Concept of reaction equilibrium, maximum attainable equilibrium extent, relationship between thermodynamics and reaction engineering, relation between equilibrium constant and Gibbs free energy, effect of pressure and temperature on equilibrium constant. (Chapter 3)
3. Material balance for chemical reactors: Batch reactor, CSTR, PFR; constant and variable density cases; multiple reactions; Comparison of CSTR & PFR performance; selection of reactor, selectivity, yield, recycle reactors etc. (Chapter 4)
4. Chemical Kinetics: (1) Molecular theory of homogenous chemical reactions; elementary reactions; quasi-steady state approximation and transition state theory. (2) Molecular theory of heterogeneous chemical reactions; physical and chemical adsorption, surface diffusion, desorption; deduction of overall kinetics (Chapter 5)
5. Analysis of catalyst pellet; design of PFR with catalyst pellets (Chapter 7)
6. Energy balance for chemical reactors (Chapter 6)
7. Mixing in chemical reactions: RTD, reactor dynamics etc. (Chapter 8)
8. Parameter estimation: (Chapter 9)

Knowledge, Abilities, and Skills Students Should Have Entering this Course:

1. This course has the following prerequisites: ChE 322, 348 and 354 with a grade of at least C.
2. A comprehensive knowledge of transport phenomena is essential. This course relies on shell balances and the constituent equations to develop the material and energy balances for the various reactors, and to develop the balances in heterogeneous catalysts that account for diffusion with reaction. Transport phenomena concepts are used to appreciate the consequences of turbulent flow on velocity, temperature and concentration profiles that form the basis of the one-dimensional models. (ChE 353)
3. Knowledge of heat transfer processes is needed for nonisothermal reactors. (ChE 353 and ChE 354)
4. Understanding of reaction equilibria and heat of reaction is required. (ChE 322)
5. The students need to be able to solve linear differential equations. (M427K)
6. The students need to know matrix multiplication. (ChE 348)
7. The students need to know how to solve ODEs using explicit numerical methods and to have a working knowledge of how these methods work. (ChE 348)
8. The students need to know how to solve sets of linear and nonlinear algebraic equations using numerical methods. (ChE 348).
9. The students need an appreciation of numerical optimization procedures. (ChE 348)
10. The students need to have a working knowledge of MATLAB numerical software that operates on the Department Learning Resource Center computers. (ChE 210 and ChE 348)

Knowledge, Abilities, and Skills Students Should Gain from this Course:

1. Ability to derive and apply the energy and material balances that are required to design isothermal and non-isothermal batch, plug flow, fixed-bed and continuous stirred tank reactors.
2. Ability to solve problems of variable density and multiple independent reactions.
3. Ability to solve problems of mass transfer with reaction in heterogeneous catalysts.
4. Ability to solve problems incorporating simple, one-parameter non-ideal reactor modeling.
5. Knowledge of chemical reaction analysis concepts and chemical kinetics concepts.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have

my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.